

2018(P)

Time : 3 hours

Full Marks : 80

Pass Marks : 32

**Candidates are required to give their answers in
their own words as far as practicable.**

The figures in the margin indicate full marks.

Answer any five questions.

- 1. What are Photoconductor ? Give a schematic diagrams of a photoconductor. Obtain expression for current gain of a photoconductor.**

4+6+6 = 16

- 2. Give the principle and working of a photo-detector.**

16

- 3. What do you mean by LED ? Discuss its uses in numeric and alphanumeric display.**

16

4. Explain internal microprocessor architectures. Discuss, with examples, the real modes and protected modes of memory paging. 16
5. Explain the different types of memories used in microprocessor. 16
6. Give the basic concepts of I/O interface. Discuss I/O address decoding. Differentiate between ADC and DAC. $4+8+4 = 16$
7. What do you understand by modular programming ? Discuss the modular programming using the keyboard and video displays. $6+10 = 16$
8. What are the different types of interrupts ? Which user faces while executing a computer programme ? Write down interrupt routine to handle divide by zero condition. $10+6 = 16$
9. Explain the functions of programmable DMK controler. 16

10. Write short notes on any two of the following :

8×2 = 16

- (a) P-N junction photodiode
- (b) Photo-transistors
- (c) RAM
- (d) Clock genator (8284 A)

